Domain and range

1 For each of the following relations:

i Find the domain.

ii Find the range.

iii Determine whether the relation is a function or not.

a $\{(8,5),(1,3),(3,4),(9,1),(2,9)\}$

b $\{(2,3),(3,8),(6,1),(8,7),(3,2)\}$

c $\{(4,4),(8,9),(2,8),(7,9),(3,1)\}$

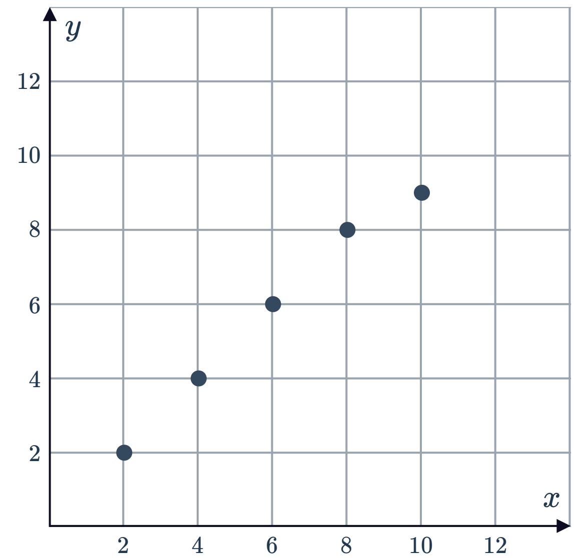
d

x	1	6	3	8	2
y	3	2	7	1	2

e

x	7	7	8	5	3
y	1	9	3	2	6

f



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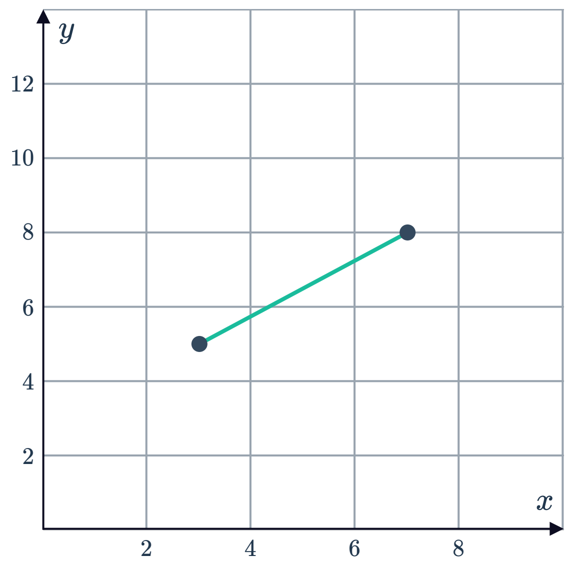

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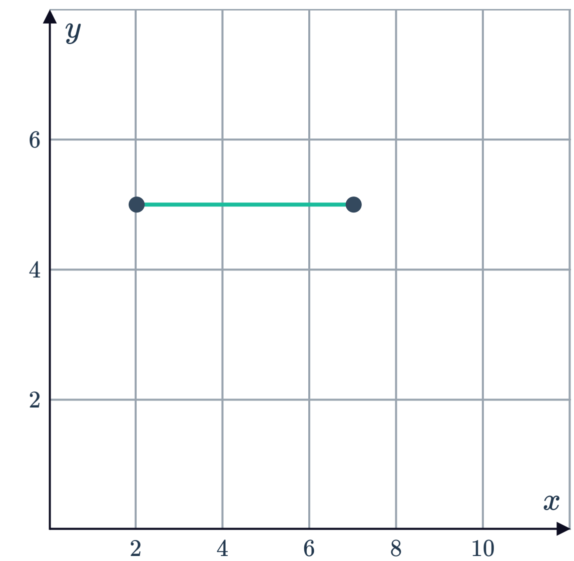
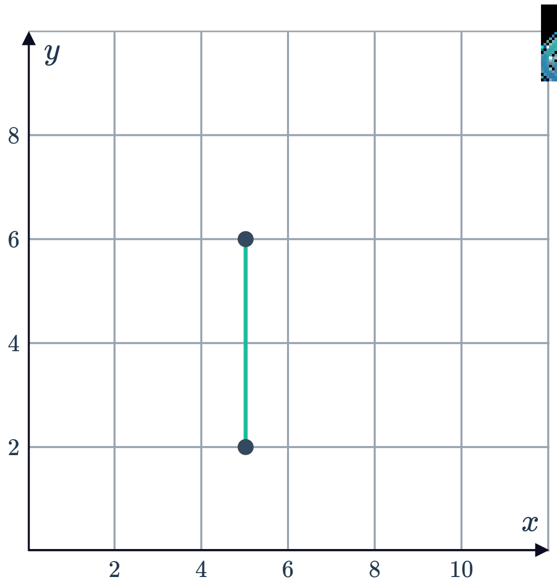
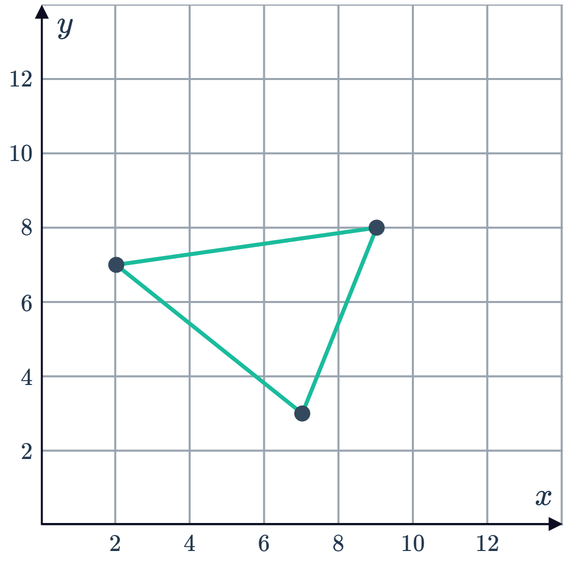


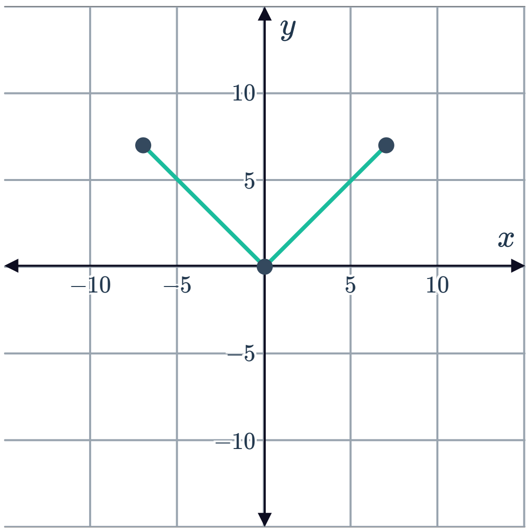
i


y
8
6
4
2
x
2 4 6 8 10

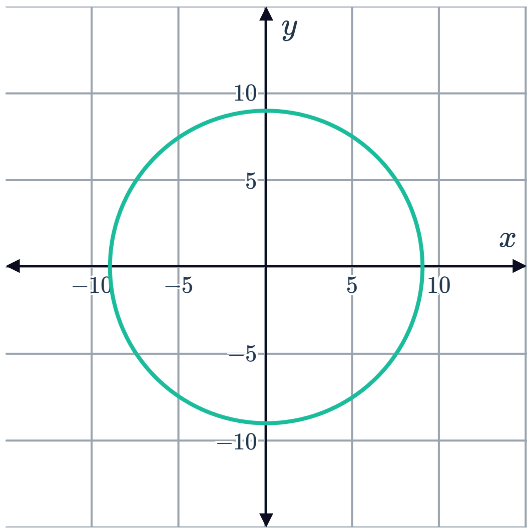
j


y
12
10
8
6
4
2
x
2 4 6 8 10 12

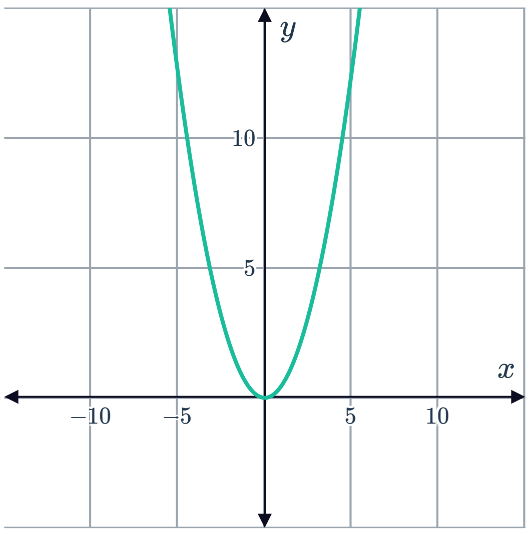
k


y
10
5
x
−10 −5 5 10
−5
−10

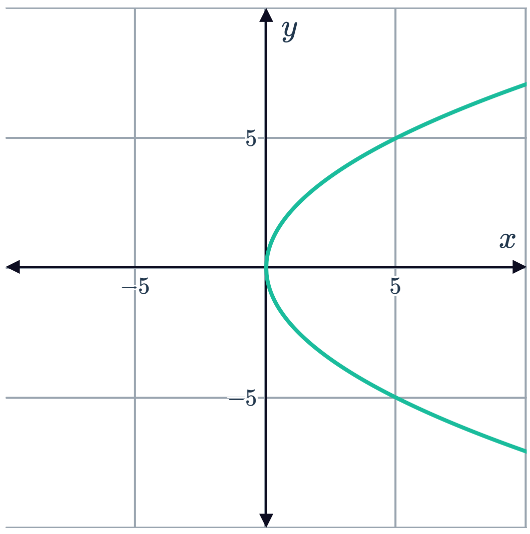
l


y
10
5
x
−10 −5 5 10
−5
−10

m

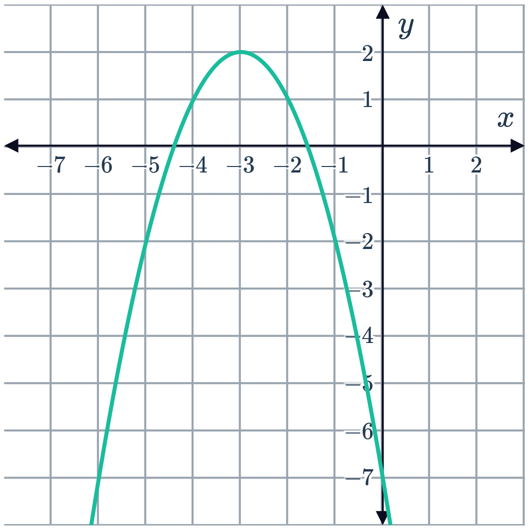

y
10
5
x
−10 −5 5 10

n


y
5
x
−5 5
−5

2 State the domain and range of the followi[redacted]nctions:

a



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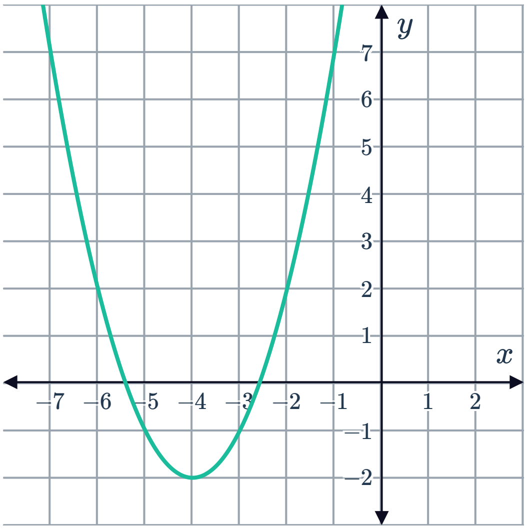
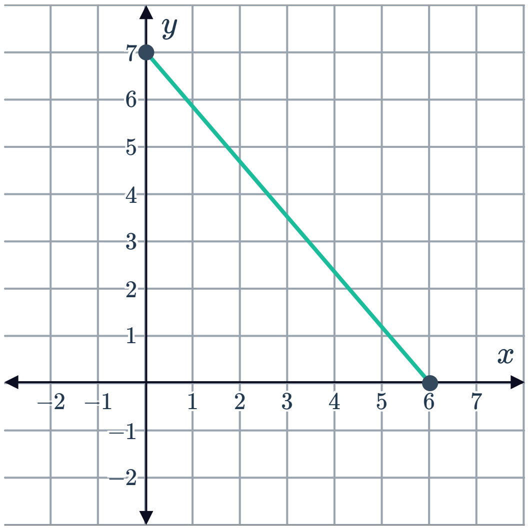
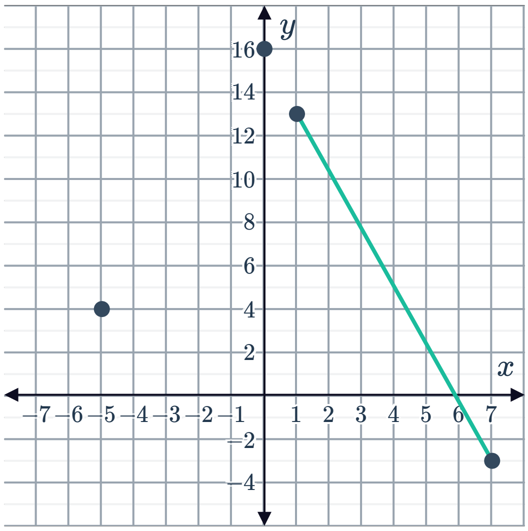


c



3 Consider the function f graphed. Use the graph to determine whether each of the given statements are true or false:



a The domain is $[-5, 7]$

b The range is $[-3, 16]$

c $f(1) - f(7) = 16$

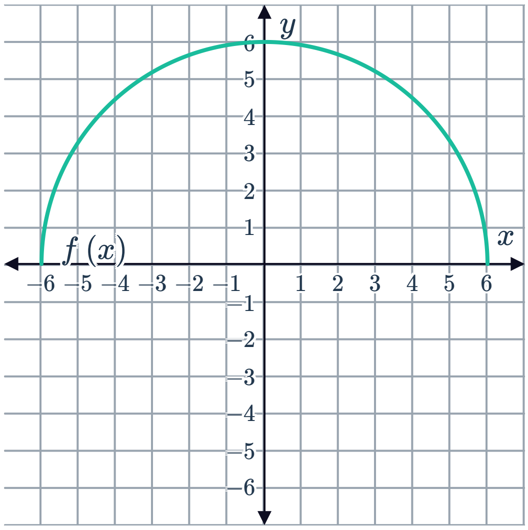
d $f(0) = 16$

4 What is the domain of the following functi

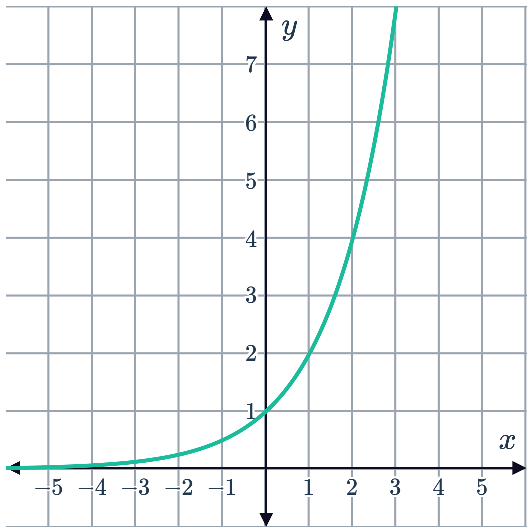
a $f(x) = -3x - 1$

b $f(x) = 8x + 8$

c



5 The graph of $y = 2^x$ is shown below:



a What is the y-intercept of this graph?

b Does the graph have an x-intercept?

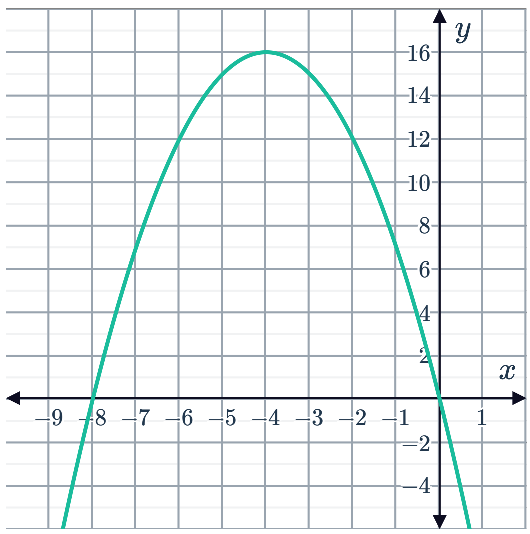
c What is the graph's domain?

d What is the graph's range?

e Find the value of y when $x = 7$.

f Find the value of x when $y = 256$.

6 Consider the graph of the function shown:

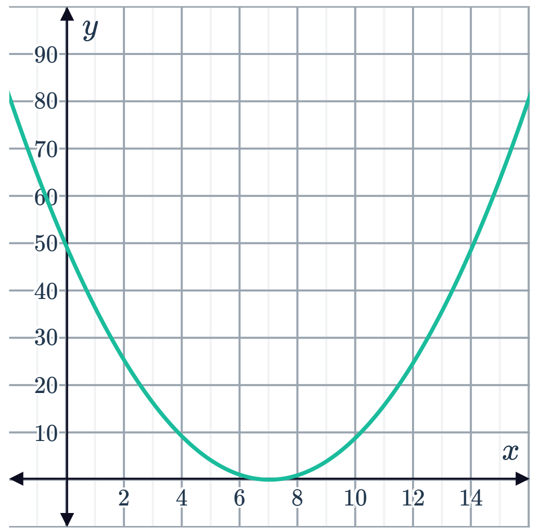


a What is the maximum value of the graph?

b Hence, determine the range of the function.

c What is the domain of this function?

7 Consider the graph of the function shown.



a What is the minimum value of the graph?

b Hence, determine the range of the function.

c Over what interval of the domain is the function increasing?

8 For the following quadratic functions:

i Determine whether the function has a maximum or a minimum point.

ii Find the minimum or maximum value of the function.

iii Find the range of the function.

a

x	y
1	17
2	7
3	1
4	-1
5	1
6	7
7	17

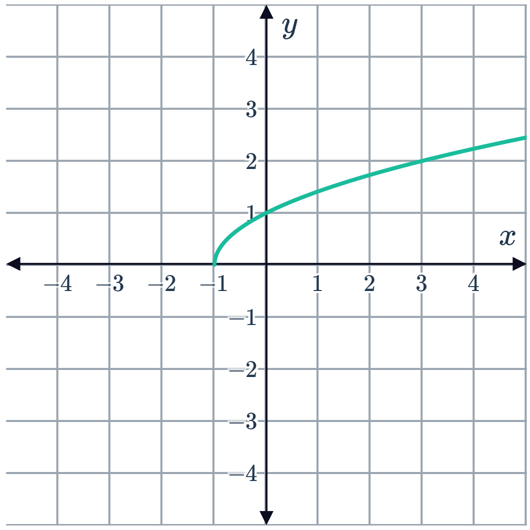
b

x	y
-5.5	-12
-5	-7
-4.5	-4
-4	-3
-3.5	-4
-3	-7
-2.5	-12

9 Consider the graph of the function $f\left(x\right) =$ [illegible] $+ 1$ below:

a State the domain of the function.

b Is there a value in the domain that can produce a function value of -2?



10 Consider the parabola defined by the equation $y = x^2 + 5$.

a Is the parabola concave up or concave down?

b What is the minimum y value of the parabola?

c Hence, determine the range of the parabola.

Graphing functions

11 For the following functions:

i Sketch the graph of the function.

ii Find the domain.

iii Find the range.

a $f\left(x\right) = -x + 1$

b $f\left(x\right) = 3x - 1$

c $f\left(x\right) = -\dfrac{3}{2}x - 2$

d $f\left(x\right) = -4$

e $x = -1$

12 Consider the function $y = x^3 - 1$.

a Complete the table of values:

x	-2	-1	0	1	2
y					

b Sketch the graph of the function.

c Is there a value in the domain that can produce a function value of $3\dfrac{2}{3}$?

13 Consider $f(x) = x + 3$ for the domain $\{-5$ [redacted] $0, 1\}$.

a Complete the table of values:

x	-5	-4	0	1
$f(x)$				

b Plot the points on a number plane.

14 Consider $f(x) = -2x$ for the domain $\{-1, 0, 1, 2\}$.

a Complete the table of values:

x	-1	0	1	2
$f(x)$				

b Plot the points on a number plane.

15 Consider the function $f(x) = \dfrac{x+3}{2}$, for all real x.

a Complete the table of values:

x	3	4	5	6	7
y					

b Sketch the graph of the function.

16 Consider $-2x + y = 1$ for the domain $\{-4, -3, 0, 2\}$.

a Make y the subject of the equation.

b Complete the table of values:

x	-4	-3	0	2
y				

c Plot the points on a number plane.

17 Consider $f(x) = x^2 + 4$ for the domain $\{-3, -1, 0, 1, 3\}$.

a Complete the table of values:

x	-3	-1	0	1	3
$f(x)$					

b Plot the points on a number plane.

18 Consider $f(x) = (x+2)^2$ for the domain [redacted]$1, 0, 1, 2\}$.

a Complete the table of values:

x	-2	-1	0	1	2
$f(x)$					

b Plot the points on a number plane.

19 The function f is used to determine the area of a square given its side length.

a State whether the following values are part of the domain of the function:

i -8 **ii** 6.5 **iii** $9\frac{1}{3}$ **iv** $\sqrt{78}$

b For $n \geq 0$, state the area function for a side length of n.

c Sketch the graph of the function f.

Rational functions

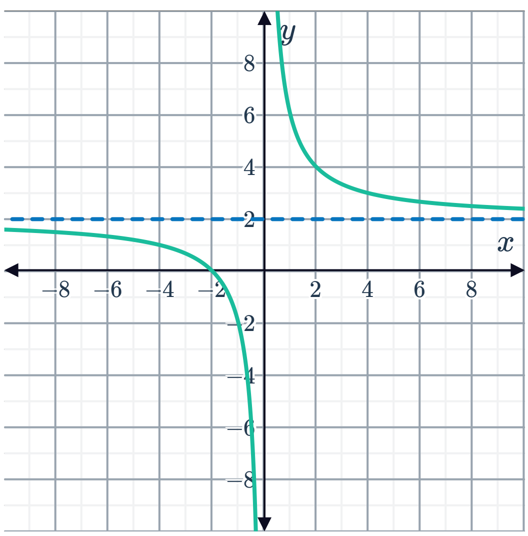
20 What is the domain of the function defined by $f(x) = \dfrac{1}{x+5}$?

21 For each of the following graphs of rational functions:

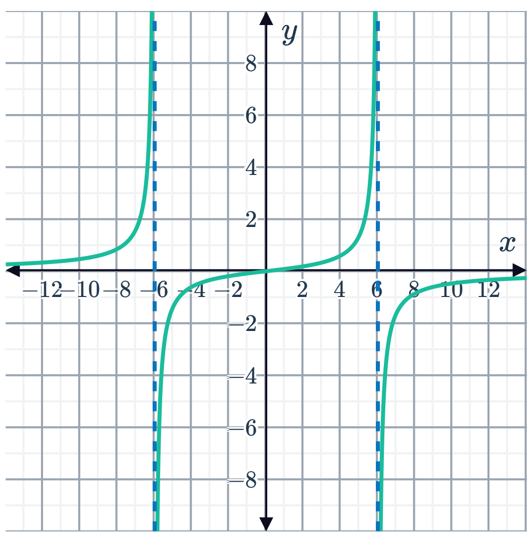
i State the domain using interval notation.

ii State the range using interval notation.

a



b



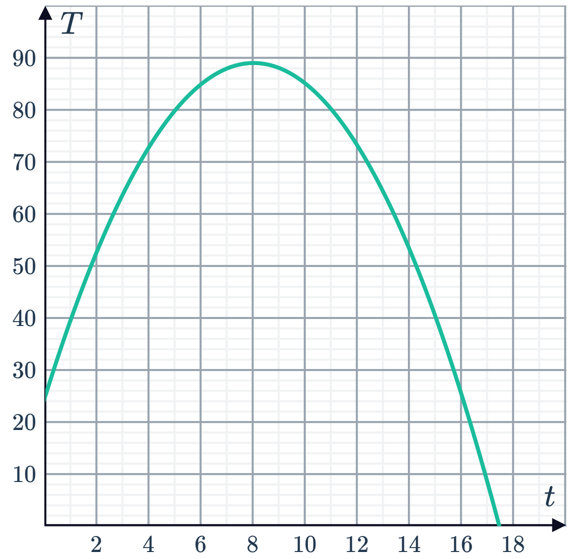
Applications

22 An industrial process involves heating a material from $25^{\circ}C$ to $80^{\circ}C$ in order to remove impurities.

Using the standard approach, the temperature of the material over time is given by the function:

$$T = -t\left(t - 16\right) + 25$$

A graph of the temperature is given, where temperature T is given in degrees Celsius $\left({}^{\circ}C\right)$ and time t is in minutes.



a What is the domain of this function?

b What is the range of this function?

c A new heating process is being developed that uses more energy but takes less time. This regime is based on a different quadratic function, and the table below shows the theoretical value of the temperature at different points in time using the new function.

t	0	1	3	4	30	31	32	33	34	35
T	25	45	80	96	96	80	63	45	25	4

For what values of t does this new function have the same range as the standard process?

d Hence, what values of t should be used for the new heating process?

23 The number of bees in a colony is initially measured and left to form a hive. The number of bees in the hive is measured each day over the course of one week. The function

$$H\left(x\right) = 30\left(3\right)^{x}$$

is found to model the number of bees, H, after x days.

a What is the domain of the function?

b What is the initial number of bees in the colony?

c Can the function be used to model the number of bees over an entire season?

24 At an indoor ski facility, the temperature is [redacted]o $-8^{\circ}C$ at the opening time of 9 am. At 10 am, the temperature is immediately brought down to $-15^{\circ}C$ and left for 3 hours before immediately taking it down again to $-23^{\circ}C$ where it stays until the close time of 7 pm.

a Write the piecewise function that models the indoor temperature $f(x)$ in terms of the number of hours after the facility has opened, x.

b The graph of the function is given. State the domain of the function.

c Kate entered the ski facility at $2{:}30$ pm. What was the temperature inside the facility?

d Vincent wants to wait until the indoor temperature is $-10^{\circ}C$ or lower. When is the earliest he can enter the facility?